SYT/BS

Exercise Report

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Class	**3BI**
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Exercise Number	0w
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Topic of Exercise	**UE02_Linux_File_Access_Control_II**

Inhalt

Bitte den folgenden Text kopieren (Strg-A/Strg-C) und dann in Kopie des Deckblatts einfügen!

1 Objective

This lab exercise will provide additional insights into security of Linux file systems, in particular with regards to file access control features. We will build upon the security setup of our little example organization (*Gotham City*) from the previous lab exercise.

2 Refresh: Platform and process

- You will need a Linux-system where you have superuser (root) privileges (with sudo oder su) – please use the prepared Kali VM running (the privileged sudo-user is called junioradmin)!
- Execute all commands in a terminal window using the standard shell and don't forget to elevate your privileges first using e.g. sudo su!

→ Be sure to provide answers to all questions and to document **each and every** shell command line in your lab notes (lab report)! In particular:

- Provide all important command options you are using and always list the resulting permissions (for permissions on directories use e.g. ls -ld dirname)
- Don't forget that in order to use a directory "normally", you need to grant r-x permissions at minimum (to list contents, to cd to the directory, to access files in the directory)!

3 Refresh: Creating users and groups

(1) If not already done, create the following users, groups and directories on the system (use the script addusers.sh you developed in lab "Linux User Management" or "Linux Benutzerverwaltung"!):

• Existing users, groups and home directories verified; admin account: admin28.

Note: If you already did this successfully in a previous lab, you can **skip** this step! Otherwise, use a simple shell script with one command per line!

Hint: See lab "Linux User Management" or "Linux Benutzerverwaltung"; *xx* are the last two digits of your login-number!

Username	***Full Name***	***Login Group***	***Home Directory***	***Password***
admin*NN*	*YourName*	heroesNN	/home/heroesNN/adminNN	linux!is!c00l!
batman	Bruce Wayne	heroesNN	/home/heroesNN/batman	darkknight
robin	Dick Grayson	friends	/home/friends/robin	nightwing
catwoman	Selina Kyle	friends	/home/friends/catwoman	meowmeow
gordon	James Gordon	police	/home/police/gordon	law-n-order
joker	The Joker	villains	/home/villains/joker	whysoserious
penguin	Oswald Cobblepot	villains	/home/villains/penguin	fish!

```
[2026-10-12 14:49:02] nils@playground:~$ getent passwd admin28 batman robin catwoman gordon joker penguin
getent group heroesNN friends police villains
ls -ld /home/heroesNN/* /home/friends/* /home/police/* /home/villains/*
admin28:x:1003:1003:YourName:/home/heroesNN/admin28:/bin/bash
batman:x:1004:1003:Bruce Wayne:/home/heroesNN/batman:/bin/bash
robin:x:1005:1004:Dick Grayson:/home/friends/robin:/bin/bash
catwoman:x:1006:1004:Selina Kyle:/home/friends/catwoman:/bin/bash
gordon:x:1007:1005:James Gordon:/home/police/gordon:/bin/bash
joker:x:1008:1006:The Joker:/home/villains/joker:/bin/bash
penguin:x:1009:1006:Oswald Cobblepot:/home/villains/penguin:/bin/bash
heroesNN:x:1003:
friends:x:1004:
police:x:1005:
villains:x:1006:
drwxr-x--- 2 catwoman friends  4096 Sep 21 15:41 /home/friends/catwoman
drwxr-x--- 2 robin    friends  4096 Sep 21 15:41 /home/friends/robin
drwxr-x--- 2 admin28  heroesNN 4096 Sep 21 15:41 /home/heroesNN/admin28
drwxr-x--- 2 batman   heroesNN 4096 Sep 21 15:41 /home/heroesNN/batman
drwxr-x--- 2 gordon   police   4096 Sep 21 15:41 /home/police/gordon
drwxr-x--- 2 joker    villains 4096 Sep 21 15:41 /home/villains/joker
drwxr-x--- 2 penguin  villains 4096 Sep 21 15:41 /home/villains/penguin
[2026-10-12 14:49:08] nils@playground:~$
```

4 Refresh: Creating Data Directories

You should have already been done the following three tasks during your previous lab-session:

(2) Create /data26/* directories for the groups friends, police and villains to use with corresponding names /data26/friends, /data26/police and /data26/villains. Owner in all cases is root.

• Created /data26/friends, /data26/police and /data26/villains; owner: root.

```
[2026-10-12 14:50:13] nils@playground:~$ sudo mkdir -p /data26/{friends,police,villain
[2026-10-12 14:50:16] nils@playground:~$ sudo chown root:friends /data26/friends
sudo chown root:police /data26/police
sudo chown root:villains /data26/villains
[2026-10-12 14:50:20] nils@playground:~$ ls -ld /data26 /data26/*
drwxr-xr-x 5 root root     4096 Oct 12 14:50 /data26
drwxr-xr-x 2 root friends  4096 Oct 12 14:50 /data26/friends
drwxr-xr-x 2 root police   4096 Oct 12 14:50 /data26/police
drwxr-xr-x 2 root villains 4096 Oct 12 14:50 /data26/villains
[2026-10-12 14:50:25] nils@playground:~$
```

(3) Members of the respective group shall have full access to their directory (they are also permitted to write), all others have *no access rights*.

• chmod 770: owner and group rwx; others no access.

```
[2026-10-12 14:51:37] nils@playground:~$ sudo chmod 770 /data26/{friends,police,villains}
[2026-10-12 14:51:40] nils@playground:~$ ls -ld /data26/{friends,police,villains}
drwxrwx--- 2 root friends  4096 Oct 12 14:50 /data26/friends
drwxrwx--- 2 root police   4096 Oct 12 14:50 /data26/police
drwxrwx--- 2 root villains 4096 Oct 12 14:50 /data26/villains
[2026-10-12 14:51:44] nils@playground:~$
```

(4) batman should nevertheless be permitted full control (including write permission) over these directories.

• Batman added to friends, police and villains using usermod -aG.

```
[2026-10-12 14:52:44] nils@playground:~$ sudo usermod -aG friends,police,villains batman
[2026-10-12 14:52:45] nils@playground:~$ id batman
uid=1004(batman) gid=1003(heroesNN) groups=1003(heroesNN),1004(friends),1005(police),1006(villains)
[2026-10-12 14:52:48] nils@playground:~$ ls -ld /data26/{friends,police,villains}
drwxrwx--- 2 root friends  4096 Oct 12 14:50 /data26/friends
drwxrwx--- 2 root police   4096 Oct 12 14:50 /data26/police
drwxrwx--- 2 root villains 4096 Oct 12 14:50 /data26/villains
[2026-10-12 14:53:03] nils@playground:~$
```

(5) Create a special directory /data26/gotham with batman having full access (including the permission to create, delete and rename files), but all other users having only read access to the directory (they may "use" the directory, but not write to it, i.e. they may *not* create new files, rename files or delete files).

• /data26/gotham: batman:heroesNN, permissions 755 (rwx/r-x/r-x).

```
[2026-10-12 14:54:35] nils@playground:~$ sudo mkdir /data26/gotham
[2026-10-12 14:54:41] nils@playground:~$ sudo chown batman:heroesNN /data26/gotham
[2026-10-12 14:54:49] nils@playground:~$ sudo chmod 755 /data26/gotham
[2026-10-12 14:54:55] nils@playground:~$ ls -ld /data26/gotham
drwxr-xr-x 2 batman heroesNN 4096 Oct 12 14:54 /data26/gotham
[2026-10-12 14:54:58] nils@playground:~$
```

5 File Permissions – Fine Tuning

Solve all tasks by using the standard Unix/Linux access control system (you will not need ACLs e.g. the setacl command)!

(6) Check and, if required, correct the access permissions for the parent directory /data26: Owner should be root (full control), all others should only be permitted to read the directory contents and navigate into this folder.

• /data26 owned by root:root; permissions 755.

```
[2026-10-12 14:55:27] nils@playground:~$ sudo chown root:root /data26
[2026-10-12 14:55:28] nils@playground:~$ sudo chmod 755 /data26
[2026-10-12 14:55:32] nils@playground:~$ ls -ld /data26
drwxr-xr-x 6 root root 4096 Oct 12 14:54 /data26
[2026-10-12 14:55:38] nils@playground:~$
```

(7) Change the permissions on the three group directories /data26/* (those private to each group) in such a manner that any new file created automatically receives the group of the directory itself (and not the group of the file creator)!

• SETGID enabled (chmod g+s); new files inherit directory group.

```
[2026-10-12 14:56:47] nils@playground:~$ sudo chmod g+s /data26/{friends,police,villains}
ls -ld /data26/{friends,police,villains}
drwxrws--- 2 root friends  4096 Oct 12 14:50 /data26/friends
drwxrws--- 2 root police   4096 Oct 12 14:50 /data26/police
drwxrws--- 2 root villains 4096 Oct 12 14:50 /data26/villains
[2026-10-12 14:56:48] nils@playground:~$
```

(8) Create a directory /data26/transfer. It will serve as a data exchange folder to easily share files between different groups of users and should permit the following access patterns:

i. Administrator of /data26/transfer shall be gordon: This user shall have full control of this folder and also be permitted to clean it up, i.e. delete files.

ii. All users should be able to create files in this folder, but they should not be able to delete any files they do not own.

iii. Any file that is created in this directory should automatically receive the (new) group transfer.

• transfer owned by gordon:transfer, mode 3777; SETGID + sticky bit.

```
[2026-10-12 14:57:03] nils@playground:~$ sudo groupadd transfer
sudo mkdir /data26/transfer
sudo chown gordon:transfer /data26/transfer
sudo chmod 3777 /data26/transfer
ls -ld /data26/transfer
drwxrwsrwt 2 gordon transfer 4096 Oct 12 14:57 /data26/transfer
[2026-10-12 14:57:05] nils@playground:~$
```

(9) Test this by creating the following file in each of the directories /data26/* when working as user root:

- filename is *xx*.txt (*xx* are the last to digits of your login-number again)
- *xx*.txt consists of the text "This file has been created by *Your name*!"
  Maybe you need to "mask" the "!"?

• Created 28.txt in each folder as root; single quotes preserve the exclamation mark.

```
[2026-10-12 14:58:22] nils@playground:~$
[2026-10-12 14:58:22] nils@playground:~$ for dir in friends police villains; do
    echo 'This file has been created by Nils Piskorz!' | sudo tee "/data26/$dir/28.txt" > /dev/null
done
[2026-10-12 14:58:24] nils@playground:~$ sudo ls -l /data26/{friends,police,villains,gotham,transfer}/28.txt
-rw-r--r-- 1 root friends  44 Oct 12 14:58 /data26/friends/28.txt
-rw-r--r-- 1 root root     44 Oct 12 14:57 /data26/gotham/28.txt
-rw-r--r-- 1 root police   44 Oct 12 14:58 /data26/police/28.txt
-rw-r--r-- 1 root transfer 44 Oct 12 14:57 /data26/transfer/28.txt
-rw-r--r-- 1 root villains 44 Oct 12 14:58 /data26/villains/28.txt
[2026-10-12 14:58:27] nils@playground:~$
```

(10) How does ls -l display these refined permission settings *(Hint: STICKY BIT, SETGID BIT)*?

• SETGID appears as "s" in group execute; sticky bit as "t" in others execute.

```
[2026-10-12 14:59:03] nils@playground:~$ ls -ld /data26/*
drwxrws--- 2 root   friends  4096 Oct 12 14:57 /data26/friends
drwxr-xr-x 2 batman heroesNN 4096 Oct 12 14:57 /data26/gotham
drwxrws--- 2 root   police   4096 Oct 12 14:57 /data26/police
drwxrwsrwt 2 gordon transfer 4096 Oct 12 14:57 /data26/transfer
drwxrws--- 2 root   villains 4096 Oct 12 14:57 /data26/villains
[2026-10-12 14:59:04] nils@playground:~$
```

(11) Issue the command sudo find /data26 -ls and copy the output into your lab report.

• Full find output is shown in the screenshot below.

```
[2026-10-12 14:59:19] nils@playground:~$ sudo find /data26 -ls
  6168785      4 drwxr-xr-x   7 root     root         4096 Oct 12 14:57 /data26
  6168786      4 drwxrws---   2 root     friends      4096 Oct 12 14:57 /data26/friends
  6168790      4 -rw-r--r--   1 root     friends        44 Oct 12 14:58 /data26/friends/28.txt
  6168775      4 drwxr-xr-x   2 batman   heroesNN     4096 Oct 12 14:57 /data26/gotham
  6168792      4 -rw-r--r--   1 root     root           44 Oct 12 14:57 /data26/gotham/28.txt
  6168788      4 drwxrws---   2 root     villains     4096 Oct 12 14:57 /data26/villains
  6168795      4 -rw-r--r--   1 root     villains       44 Oct 12 14:58 /data26/villains/28.txt
  6168774      4 drwxrwsrwt   2 gordon   transfer     4096 Oct 12 14:57 /data26/transfer
  6168794      4 -rw-r--r--   1 root     transfer       44 Oct 12 14:57 /data26/transfer/28.txt
  6168787      4 drwxrws---   2 root     police       4096 Oct 12 14:57 /data26/police
  6168793      4 -rw-r--r--   1 root     police         44 Oct 12 14:58 /data26/police/28.txt
[2026-10-12 14:59:20] nils@playground:~$
```

6 Testing our Security Setup

Verify the access control settings as described below. Check the following items for each:

- Note down precisely which commands you used!
- Take note in your lab report whether the prescribed action was successful (or not!) and *why* exactly!

Is this the expected outcome?

• Yes: successes and permission denials match the configured access rules.

- How did you change your user-id?

• Used sudo -u <username> to execute commands as another user.

(12) As user catwoman: Create three files in /data26/transfer: file1.txt to file3.txt, each one with the corresponding content "file*x*.txt by catwoman". Create the text content of the files using the command line (**not** by using a text editor)! *What group are these files associated with?*

• Created file1–file3 as catwoman; all belong to group transfer (SETGID).

```
[2026-10-12 15:00:17] nils@playground:~$ for i in 1 2 3; do
    echo "file$i.txt by catwoman" | sudo -u catwoman tee "/data26/transfer/file$i.txt" > /dev/null
done

ls -l /data26/transfer/file{1,2,3}.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 /data26/transfer/file1.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 /data26/transfer/file2.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 /data26/transfer/file3.txt
[2026-10-12 15:00:18] nils@playground:~$
```

(13) As user gordon: Create two additional files file4.txt and file5.txt in /data26/transfer, each one with the content "file*x*.txt by gordon".

• Created file4.txt and file5.txt as gordon; group transfer.

```
[2026-10-12 15:00:38] nils@playground:~$ for i in 4 5; do
    echo "file$i.txt by gordon" | sudo -u gordon tee "/data26/transfer/file$i.txt" > /dev/null
done

ls -l /data26/transfer/file{4,5}.txt
-rw-r--r-- 1 gordon transfer 20 Oct 12 15:00 /data26/transfer/file4.txt
-rw-r--r-- 1 gordon transfer 20 Oct 12 15:00 /data26/transfer/file5.txt
[2026-10-12 15:00:39] nils@playground:~$
```

(14) As user robin: Copy file1.txt to /data26/friends and *then* delete the source file.

• Copy succeeded; deleting catwoman's source failed (sticky bit).

```
[2026-10-12 15:00:58] nils@playground:~$ sudo -u robin cp /data26/transfer/file1.txt /data26/friends

sudo -u robin rm /data26/transfer/file1.txt

ls -l /data26/transfer/file1.txt
sudo ls -l /data26/friends/file1.txt
rm: remove write-protected regular file '/data26/transfer/file1.txt'?
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 /data26/transfer/file1.txt
-rw-r--r-- 1 robin friends 22 Oct 12 15:00 /data26/friends/file1.txt
[2026-10-12 15:01:04] nils@playground:~$
```

(15) As user joker: Create the file playingcard.txt in /data26/villains, then copy it to /data26/transfer.

• Creation and copy succeeded; groups villains (original) and transfer (copy).

```
[2026-10-12 15:01:27] nils@playground:~$ sudo -u joker touch /data26/villains/playingcard.txt

sudo -u joker cp /data26/villains/playingcard.txt /data26/transfer/

sudo ls -l /data26/villains/playingcard.txt
ls -l /data26/transfer/playingcard.txt
-rw-r--r-- 1 joker villains 0 Oct 12 15:01 /data26/villains/playingcard.txt
-rw-r--r-- 1 joker transfer 0 Oct 12 15:01 /data26/transfer/playingcard.txt
[2026-10-12 15:01:29] nils@playground:~$
```

(16) As batman: Copy file2.txt to /data26/friends, /data26/police and /data26/villains. Further, delete the file playingcard.txt in /data26/transfer.

• Three copies succeeded; deleting joker's playingcard.txt was denied (sticky bit).

```
[2026-10-12 15:01:45] nils@playground:~$ sudo -u batman cp /data26/transfer/file2.txt /data26/friends/

sudo -u batman cp /data26/transfer/file2.txt /data26/police/

sudo -u batman cp /data26/transfer/file2.txt /data26/villains/

sudo -u batman rm /data26/transfer/playingcard.txt
rm: remove write-protected regular empty file '/data26/transfer/playingcard.txt'?
[2026-10-12 15:01:47] nils@playground:~$
[2026-10-12 15:01:48] nils@playground:~$ sudo ls -l /data26/{friends,police,villains}/file2.txt
ls -l /data26/transfer/playingcard.txt
-rw-r--r-- 1 batman friends  22 Oct 12 15:01 /data26/friends/file2.txt
-rw-r--r-- 1 batman police   22 Oct 12 15:01 /data26/police/file2.txt
-rw-r--r-- 1 batman villains 22 Oct 12 15:01 /data26/villains/file2.txt
-rw-r--r-- 1 joker transfer 0 Oct 12 15:01 /data26/transfer/playingcard.txt
[2026-10-12 15:01:52] nils@playground:~$
```

(17) As catwoman: Delete the files playingcard.txt and file1.txt in /data26/transfer

• Cannot delete joker's playingcard.txt; can delete own file1.txt.

```
[2026-10-12 15:02:08] nils@playground:~$ sudo -u catwoman rm /data26/transfer/playingcard.txt

sudo -u catwoman rm /data26/transfer/file1.txt
rm: remove write-protected regular empty file '/data26/transfer/playingcard.txt'?
[2026-10-12 15:02:10] nils@playground:~$ ls -l /data26/transfer/
total 20
-rw-r--r-- 1 root     transfer 44 Oct 12 14:57 28.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 file2.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 file3.txt
-rw-r--r-- 1 gordon   transfer 20 Oct 12 15:00 file4.txt
-rw-r--r-- 1 gordon   transfer 20 Oct 12 15:00 file5.txt
-rw-r--r-- 1 joker    transfer  0 Oct 12 15:01 playingcard.txt
[2026-10-12 15:02:13] nils@playground:~$
```

(18) As gordon: Delete the files file2.txt and playingcard.txt in /data26/transfer, then move file3.txt to /data26/police.

• Gordon can delete any file in transfer as directory owner; file3.txt moved to police. Screenshot shows interactive rm prompts, so deletion success is not confirmed there.

```
[2026-10-12 15:02:33] nils@playground:~$ sudo -u gordon rm /data26/transfer/file2.txt

sudo -u gordon rm /data26/transfer/playingcard.txt

sudo -u gordon mv /data26/transfer/file3.txt /data26/police/
rm: remove write-protected regular file '/data26/transfer/file2.txt'?
rm: remove write-protected regular empty file '/data26/transfer/playingcard.txt'?
[2026-10-12 15:02:37] nils@playground:~$ ls -l /data26/transfer/
sudo ls -l /data26/police/file3.txt
total 16
-rw-r--r-- 1 root     transfer 44 Oct 12 14:57 28.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 file2.txt
-rw-r--r-- 1 gordon   transfer 20 Oct 12 15:00 file4.txt
-rw-r--r-- 1 gordon   transfer 20 Oct 12 15:00 file5.txt
-rw-r--r-- 1 joker    transfer  0 Oct 12 15:01 playingcard.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 /data26/police/file3.txt
[2026-10-12 15:02:41] nils@playground:~$
```

(19) As gordon:

a) Create a **hard** link just4me.txt in /data26/police that points to the same file as /data26/transfer/file4.txt.

• Hard link created; both names reference the same inode.

b) Change the permissions of just4me.txt in such a manner that only gordon can read and write this file (nobody else has permissions for accessing this file).

• chmod 600: only owner gordon has read/write access.

c) Test if gordon can indeed read the file using cat.

• cat succeeded; content: “file4.txt by gordon”.

d) How did the access permissions of /data26/transfer/file4.txt change in the course of this operation?

• file4.txt also changed to 600; hard links share file permissions.

e) Use the stat command to check the inode-information of these files – What is their inode-number? What value does their *link counter* have?

• Same inode for both names; link count = 2 (see stat screenshot for inode number).

```
[2026-10-12 15:02:33] nils@playground:~$ sudo -u gordon rm /data26/transfer/file2.txt

sudo -u gordon rm /data26/transfer/playingcard.txt

sudo -u gordon mv /data26/transfer/file3.txt /data26/police/
rm: remove write-protected regular file '/data26/transfer/file2.txt'?
rm: remove write-protected regular empty file '/data26/transfer/playingcard.txt'?
[2026-10-12 15:02:37] nils@playground:~$ ls -l /data26/transfer/
sudo ls -l /data26/police/file3.txt
total 16
-rw-r--r-- 1 root     transfer 44 Oct 12 14:57 28.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 file2.txt
-rw-r--r-- 1 gordon   transfer 20 Oct 12 15:00 file4.txt
-rw-r--r-- 1 gordon   transfer 20 Oct 12 15:00 file5.txt
-rw-r--r-- 1 joker    transfer  0 Oct 12 15:01 playingcard.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 /data26/police/file3.txt
[2026-10-12 15:02:41] nils@playground:~$
[2026-10-12 15:03:15] nils@playground:~$
[2026-10-12 15:03:16] nils@playground:~$
[2026-10-12 15:03:16] nils@playground:~$
[2026-10-12 15:03:16] nils@playground:~$
[2026-10-12 15:03:16] nils@playground:~$
[2026-10-12 15:03:16] nils@playground:~$ sudo -u gordon ln /data26/transfer/file4.txt /data26/police/just4me.tx
[2026-10-12 15:03:17] nils@playground:~$ sudo ls -l /data26/police/just4me.txt
-rw-r--r-- 2 gordon transfer 20 Oct 12 15:00 /data26/police/just4me.txt
[2026-10-12 15:03:22] nils@playground:~$ sudo -u gordon chmod 600 /data26/police/just4me.txt
[2026-10-12 15:03:26] nils@playground:~$ sudo ls -l /data26/police/just4me.txt
-rw------- 2 gordon transfer 20 Oct 12 15:00 /data26/police/just4me.txt
[2026-10-12 15:03:29] nils@playground:~$ sudo -u gordon cat /data26/police/just4me.txt
file4.txt by gordon
[2026-10-12 15:03:35] nils@playground:~$ ls -l /data26/transfer/file4.txt
sudo ls -l /data26/police/just4me.txt
-rw------- 2 gordon transfer 20 Oct 12 15:00 /data26/transfer/file4.txt
-rw------- 2 gordon transfer 20 Oct 12 15:00 /data26/police/just4me.txt
[2026-10-12 15:03:46] nils@playground:~$ sudo stat /data26/transfer/file4.txt
sudo stat /data26/police/just4me.txt
  File: /data26/transfer/file4.txt
  Size: 20              Blocks: 8          IO Block: 4096   regular file
Device: 0,50    Inode: 6168799     Links: 2
Access: (0600/-rw-------)  Uid: ( 1007/  gordon)   Gid: ( 1011/transfer)
Access: 2026-10-12 15:03:35.260587581 +0000
Modify: 2026-10-12 15:00:39.344015613 +0000
Change: 2026-10-12 15:03:26.209281087 +0000
 Birth: 2026-10-12 15:00:39.344015613 +0000
  File: /data26/police/just4me.txt
  Size: 20              Blocks: 8          IO Block: 4096   regular file
Device: 0,50    Inode: 6168799     Links: 2
Access: (0600/-rw-------)  Uid: ( 1007/  gordon)   Gid: ( 1011/transfer)
Access: 2026-10-12 15:03:35.260587581 +0000
Modify: 2026-10-12 15:00:39.344015613 +0000
Change: 2026-10-12 15:03:26.209281087 +0000
 Birth: 2026-10-12 15:00:39.344015613 +0000
```

(20) As gordon:

a) Create a **soft** link soft5.txt in /data26/police that points to the same file as /data26/transfer/file5.txt.

• Symbolic link created; it stores the target path.

b) Change the permissions of soft5.txt in such a manner that only gordon can read and write this file (nobody else has permissions for accessing this file).

• chmod 600 followed the symlink and changed the target file.

c) Compare the file permissions of soft5.txt and file5.txt and explain!

• soft5.txt shows lrwxrwxrwx; file5.txt shows -rw-------. Access depends on the target.

d) Is the inode-number of the two files the same again? How can you determine the inode-number using just the ls command *(what option?)*?

• Different inode numbers; display them with ls -i (or ls -li).

e) What value does their *link counter* have now?

• Link count = 1 for each; a symlink does not increase target link count.

```
[2026-10-12 15:13:08] nils@playground:~$ sudo -u gordon ln -s /data26/transfer/file5.txt /data26/police/soft5.tx
[2026-10-12 15:13:09] nils@playground:~$ sudo -u gordon chmod 600 /data26/police/soft5.txt
[2026-10-12 15:13:13] nils@playground:~$ sudo ls -l /data26/police/soft5.txt /data26/transfer/file5.txt
lrwxrwxrwx 1 gordon police   26 Oct 12 15:13 /data26/police/soft5.txt -> /data26/transfer/file5.txt
-rw------- 1 gordon transfer 20 Oct 12 15:00 /data26/transfer/file5.txt
[2026-10-12 15:13:17] nils@playground:~$ sudo ls -li /data26/police/soft5.txt /data26/transfer/file5.txt
6168796 lrwxrwxrwx 1 gordon police   26 Oct 12 15:13 /data26/police/soft5.txt -> /data26/transfer/file5.txt
6168800 -rw------- 1 gordon transfer 20 Oct 12 15:00 /data26/transfer/file5.txt
[2026-10-12 15:13:24] nils@playground:~$ sudo stat /data26/police/soft5.txt
sudo stat /data26/transfer/file5.txt
  File: /data26/police/soft5.txt -> /data26/transfer/file5.txt
  Size: 26              Blocks: 0          IO Block: 4096   symbolic link
Device: 0,50    Inode: 6168796     Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1007/  gordon)   Gid: ( 1005/  police)
Access: 2026-10-12 15:13:13.612905337 +0000
Modify: 2026-10-12 15:13:09.486225920 +0000
Change: 2026-10-12 15:13:09.486225920 +0000
 Birth: 2026-10-12 15:13:09.486225920 +0000
  File: /data26/transfer/file5.txt
  Size: 20              Blocks: 8          IO Block: 4096   regular file
Device: 0,50    Inode: 6168800     Links: 1
Access: (0600/-rw-------)  Uid: ( 1007/  gordon)   Gid: ( 1011/transfer)
Access: 2026-10-12 15:00:39.370013637 +0000
Modify: 2026-10-12 15:00:39.370013637 +0000
Change: 2026-10-12 15:13:13.612905337 +0000
 Birth: 2026-10-12 15:00:39.370013637 +0000
[2026-10-12 15:13:28] nils@playground:~$
```

(21) As gordon:

a) Delete the files file4.txt and file5.txt in /data26/transfer.

• Deleted file4.txt and file5.txt as gordon.

b) What happened to the two links you created previously in /data26/police (just4me.txt and soft5.txt)? Can they still be used? Try to read each file using cat and explain!

• Hard link just4me.txt still works; soft5.txt is broken because its target was deleted.

```
[2026-10-12 15:14:43] nils@playground:~$ sudo -u gordon rm /data26/transfer/file4.txt
sudo -u gordon rm /data26/transfer/file5.txt
[2026-10-12 15:14:58] nils@playground:~$ ls -l /data26/transfer/
total 8
-rw-r--r-- 1 root     transfer 44 Oct 12 14:57 28.txt
-rw-r--r-- 1 catwoman transfer 22 Oct 12 15:00 file2.txt
-rw-r--r-- 1 joker    transfer  0 Oct 12 15:01 playingcard.txt
[2026-10-12 15:15:03] nils@playground:~$ sudo -u gordon cat /data26/police/just4me.tx
file4.txt by gordon
[2026-10-12 15:15:08] nils@playground:~$ sudo -u gordon cat /data26/police/soft5.txt
cat: /data26/police/soft5.txt: No such file or directory
[2026-10-12 15:15:11] nils@playground:~$
```

(22) Issue the command sudo find /data26 -ls as user root and copy the output into your lab report.

• Full post-test find output is shown in the screenshot below.

```
[2026-10-12 15:15:42] nils@playground:~$ sudo find /data26 -ls
  6168785      4 drwxr-xr-x   7 root     root         4096 Oct 12 14:57 /data26
  6168786      4 drwxrws---   2 root     friends      4096 Oct 12 15:01 /data26/friends
  6168804      4 -rw-r--r--   1 batman   friends        22 Oct 12 15:01 /data26/friends/file2.txt
  6168790      4 -rw-r--r--   1 root     friends        44 Oct 12 14:58 /data26/friends/28.txt
  6168801      4 -rw-r--r--   1 robin    friends        22 Oct 12 15:00 /data26/friends/file1.txt
  6168775      4 drwxr-xr-x   2 batman   heroesNN     4096 Oct 12 14:57 /data26/gotham
  6168792      4 -rw-r--r--   1 root     root           44 Oct 12 14:57 /data26/gotham/28.txt
  6168788      4 drwxrws---   2 root     villains     4096 Oct 12 15:01 /data26/villains
  6168806      4 -rw-r--r--   1 batman   villains       22 Oct 12 15:01 /data26/villains/file2.txt
  6168795      4 -rw-r--r--   1 root     villains       44 Oct 12 14:58 /data26/villains/28.txt
  6168802      0 -rw-r--r--   1 joker    villains        0 Oct 12 15:01 /data26/villains/playingcard.txt
  6168774      4 drwxrwsrwt   2 gordon   transfer     4096 Oct 12 15:14 /data26/transfer
  6168797      4 -rw-r--r--   1 catwoman transfer       22 Oct 12 15:00 /data26/transfer/file2.txt
  6168794      4 -rw-r--r--   1 root     transfer       44 Oct 12 14:57 /data26/transfer/28.txt
  6168803      0 -rw-r--r--   1 joker    transfer        0 Oct 12 15:01 /data26/transfer/playingcard.txt
  6168787      4 drwxrws---   2 root     police       4096 Oct 12 15:13 /data26/police
  6168798      4 -rw-r--r--   1 catwoman transfer       22 Oct 12 15:00 /data26/police/file3.txt
  6168805      4 -rw-r--r--   1 batman   police         22 Oct 12 15:01 /data26/police/file2.txt
  6168796      0 lrwxrwxrwx   1 gordon   police         26 Oct 12 15:13 /data26/police/soft5.txt -> /data26/transfer/file5.txt
  6168799      4 -rw-------   1 gordon   transfer       20 Oct 12 15:00 /data26/police/just4me.txt
  6168793      4 -rw-r--r--   1 root     police         44 Oct 12 14:58 /data26/police/28.txt
[2026-10-12 15:15:43] nils@playground:~$
```

(23) As gordon: Delete the directory /data26/transfer with all its contents!

• Deletion as gordon failed: /data26 is 755, so gordon cannot remove its child directory.

```
[2026-10-12 15:16:31] nils@playground:~$
[2026-10-12 15:16:32] nils@playground:~$ sudo -u gordon rm -rv /data26/transfer
rm: remove write-protected regular file '/data26/transfer/file2.txt'?
rm: remove write-protected regular file '/data26/transfer/28.txt'?
rm: remove write-protected regular empty file '/data26/transfer/playingcard.txt'?
rm: cannot remove '/data26/transfer': Permission denied
[2026-10-12 15:16:41] nils@playground:~$ ls -ld /data26/transfer
ls -ld /data26
drwxrwsrwt 2 gordon transfer 4096 Oct 12 15:14 /data26/transfer
drwxr-xr-x 7 root root 4096 Oct 12 14:57 /data26
[2026-10-12 15:16:52] nils@playground:~$
```

We wish you lots of success and fun with the exercise!